Blur-ASKNet: An Effective Network for Remote Sensing Image Classification Capturing Joint Features

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***Abstract*—Remote sensing image classification faces two major challenges: the intra-class variability and inter-class similarity. Consequently, it is more challenging for classification neural networks to extract discriminant features. While many studies have proposed solutions, they struggle to balance model performance with discriminant feature extraction. In this paper, we propose the Blur-ASKNet. The network can effectively capture both local and global features, referred to as joint features. The network employs blurpooling to handle intra-class variability, incorporates spatial and channel attention mechanisms for multiscale adaptive feature fusion, and introduces ASKM to extract joint features at both local and global levels. Additionally, it can be deployed on small and medium-sized systems to address inter-class similarity. We conducted comparative and ablation experiments on four typical remote sensing datasets to explore the differences between remote sensing and traditional images and evaluate individual modules of the proposed model. The model performance was subsequently measured. Experimental results demonstrate that the network achieved higher recognition accuracy than the state-of-the-art methods, while still maintaining acceptable complexity.**

Index Terms*—Remote Sensing Image Classification; Neural Network; Blurpooling; Joint Features Extraction

I. INTRODUCTION

Remote sensing image classification has been widely used in many fields such as land planning, natural disaster observation, and environmental protection, and has been a very important research direction in the field of remote sensing. Remote sensing images are not exactly the same as natural images, and one of the major challenges in their processing is the presence of natural intra-class variability and inter-class similarity. The inter-class similarity represents that the classification object has more details and variations within the object whose overall difference is not significant, so it puts a high demand on the ability of capturing local features. And intra-class variaty means that different classes of images have some similarity, such as mountain and forest. This puts a higher demand on the overall feature information acquisition.

To solve this problem, previous work has used attention mechanisms to capture key features in remote sensing images.

For example, Tong et.al. [31]incorporates channel attention mechanisms in Densenet networks to obtain key features of images quickly. Meng et.al. [24] propose to an RSAM residual space attention module to capture the key spatial regions of multilevel feature maps and combine these key regions with the final feature maps as features for classification. Li et.al. [20] propose a spatial plus channel attention mechanism used on multiple convolutional feature maps to obtain a semantic representation, according to which the feature maps of different layers are guided for feature aggregation. These methods use either the channel attention mechanism or the spatial attention mechanism or a combination of both to capture the key local features, but they are not able to extract the global features of the image well.

Then, Transformer [32] came into being, and the attention mechanism of the Transformer exhibits superior capability in capturing global contextual feature information. Hence, it offers certain advantages in remote sensing image classification at the scene level. Although Transformer for scene recognition can extract contextual information well for scene level recognition, and because Transformer focuses more on the relationship between patches and ignores the features inside the patches, the extraction of local features still has defects. Many methods have been improved this year, Deng et.al. [4]proposed a joint framework of CNN-Transformer to capture more comprehensive features. zhang et.al. [37] proposed a TRS architecture, i.e., using the self-attention layer of Transformer to replace the convolution operation of Block in resnet. Qian et.al. [28] proposed a multi-instance Vision Transformer, which combines the MIL formulation and Vision Transformer,thus enhancing the representation of local features. Some of the above improvements do enable the model to capture global features and local features, but they all improve the model while retaining the main structure of Vision Transformer. However,Vision Transformer's primary structure and training methods determine the high computational cost of the model, which limits deployment in satellite or small to medium-sized equipment systems.

To address the above problems, we propose a multiscale deep network based on hybrid attention mechanisms to solve the above problems. By combining multiple attention mechanisms and the advantage of convolutional pooling with translation invariance, the model can obtain a comprehensive feature representation with a small increase in the number of parameters compared to the classical CNN network. First, in order to ensure that our network model is translation invariant, we add blurpool convolutional pooling [38] at the beginning of the model to make the model location translation invariant. In addition, we use the efficient channel attention mechanism ECA [33] to weight the generated large number of channel feature maps, ECA makes the model adaptively learn the weight of the feature maps, the

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effective feature maps will be weighted more, and the ineffective or small effect feature maps will be weighted less, so that the model can extract more critical deep features. Meanwhile, in order to extract local and global features in depth features in an integrated manner, we propose a ASKM(Attentive Selected Kernel Module) module that combines a spatial attention mechanism and a channel attention mechanism based on SK convolution, where the field of perception (RF) of SK convolution is variable and different. Size of convolution kernels can be selected to obtain multiscale features of images adaptively and obtain adequate global feature representation by channel attention weighting. While the spatial attention mechanism [17]can obtain key feature representations of local regions of interest, the ASKM module extracts both local features and global features effectively to obtain a comprehensive feature representation for subsequent classification.

The contributions of this paper can be summarized as follows:

- For the large number of channels generated duringthe feature extraction of Resnext, we add efficient attention mechanism after the channels to extract the channel feature maps of interest, and add a blurpol convolutional pooling layer to the model to make the model translation invariant.
- We design a ASK module that combines a spatial attention mechanism with a variable convolutional channel attention mechanism to extract important local features using spatial attention while also obtaining global features of the image by multi-scale convolutional channel attention, making it possible to obtain an effective joint representation of local features and global features with a small increase in parameters.
- We conducted extensive experiments on four populardatasets to validate the effectiveness of the proposed model. The results show that our approach has superior performance with small number of parameters compared to the current mainstream models.

II. Related Works

In the field of computer vision, people generally classify by the extracted features. In the early days, people classified by the extracted handcrafted features, for example, Li et.al. [19] use color histogram for remote sensing image classification, and Lowe et.al. [23] extract invariant features in the images to enable objects to be recognized well even under large scale affine transformation. Zhang et.al. [36] propose a model to extract computable texture information features from images and classify them by extracting unique texture features. Navneet et.al. [2] proposes a gradient direction histogram, which can describe local object features well by gradient and edge direction distribution. There are also some other methods relied on handcrafted features, e.g., [11], [15], [16], [18], [25], [26], [35]. However, the extracted manual features still have limitations and cannot reflect the deeper information of the image well.

With the development in the field of deep learning, the effectiveness of convolutional neural networks for deep feature extraction was discovered, leading to their application in remote sensing image scene classification, where Han et.al. [9] proposed an improved architecture of pre-trained Alexnet, combining spatial pyramidal pooling and side supervision to obtain good performance, Shafaey et.al. [29] used resnet network to extract deep features of remote sensing images to obtain deep and effective semantic representations of various scenes. Othman et.al. [27] propose a model that combines convolutional features and sparse autoencoders to learn the representation information. Lin et.al. [22] first propose an unsupervised model based on adversarial generative networks with multilayer representation for remote sensing image classification.

In recent years, researchers proposed attention mechanisms for image classification to extract key features in images such as SEnet [14], STN [17], SKnet [21] and so on. Because of its superior performance, the attention mechanism has been applied to the field of RSSC afterwards. For example, Tong et.al. [31] incorporates the channel attention mechanism in the Densenet network to obtain the image key features quickly. Meng et.al. [24] propose a residual space attention module of RSAM to capture key spatial regions of multi-level feature maps and combine these key regions with the final feature maps as features for classification. Li et.al. [20] use a spatial plus channel attention mechanism on multiple convolutional feature maps to obtain a semantic representation, according to which the feature maps of various layers are utilized to facilitate feature aggregation.

Transformer has good extraction effect on global features and is capable of capturing long-term dependencies, and likewise variant, Vision Transformer [7], also has good performance in the field of image classification. So it can also be applied to the field of remote sensing image classification. However, the Transformer pays attention to global features and does not have the advantage of processing ability of local features within patches. Therefore, Deng et.al. [5] propose a joint CNN-Transformer framework, using both Transformer stream and CNN stream to train the model simultaneously using a joint loss function. Zhang et.al. [37] propose a structure of TRS, a self-attention layer of Transformer is used to replace the convolution operation of Block in resnet to improve the representation learning ability of the model. Qian et.al. propose a multi-instance Vision Transformer [28], which combines the MIL formulation and Vision Transformer, thus enhancing the representation of local features. However, these improvements, are usually still based on the main body of the Transformer for improvement, and still suffer from the problems of large number of parameters, difficulty in deployment for small and medium-sized devices. Our model can achieve effective extraction of local features and global features through the combination of lightweight attention modules, and achieves robust model performance with low number of parameters.

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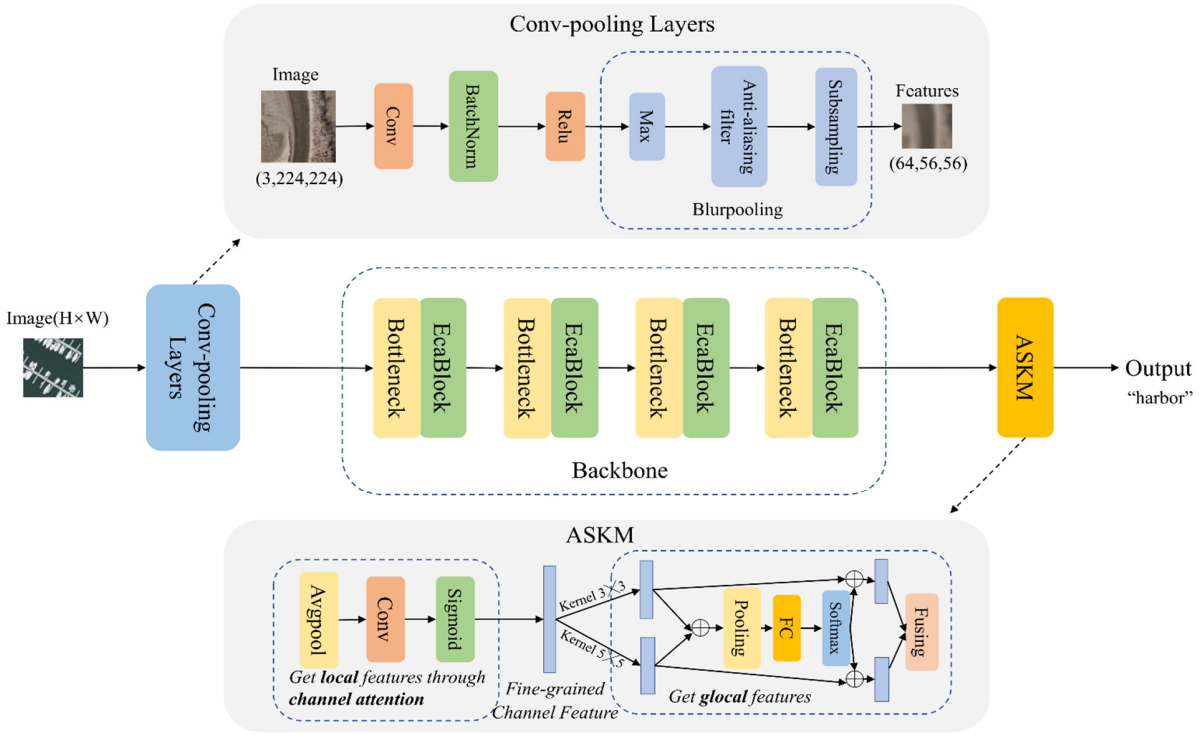


Fig. 1. The Overall Architecture of Our Proposed Model named Blur-ASKNet. The whole model is integrated with three parts, the convolution-pooling layers,the backbone model and the ASKM.

III. METHODS

The overall structure of our model is shown in Figure 1.

The flow diagram of convolution-pooling layers is shown in Figure 1.We set the kernel size of the convolution layer to 3, stride to 2, and padding to 3. The number of channels in the Batch Norm layer is set to 64. It is worth mentioning that after the convolution layer, we do not use the traditional methods of max pooling, average pooling, etc., but use blurpooling [38]. For the possible blurring, noise variants, etc. in remote sensing images, blurpooling can ensure better robustness and accuracy of the recognition model. blur pooling method finds the shift equivariance/invariance property in convolution, pooling, for images X, the shifts of H and W dimensions are set to Δh and Δw, and the shift quantity S satisfies:

$$[S_{\Delta h,\Delta w}(X)]_{h,w,c} = X_{(h-\Delta h)\%H,(w-\Delta w)\%W,c} \tag{1}$$

The % in the above equation represents the modulus function and c represents the number of channels of the image. Based on this property, blurpooling works. Blurpooling follows two major steps, one is the observation of shift equivariance and the other is subsampling. combining this property, we choose the filters with scales [1,4,4,1] respectively, reducing the aliasing.

A. *Backbone with Efficient Channel Attention*

In the structure shown in Figure 1, it can be seen that Backbone mainly consists of Bottleneck, referring to the structure in [12], lowering and raising the feature dimension respectively, the main purpose is to reduce the number of parameters, thus reducing the amount of computation, and after the dimensionality reduction can be more effective and The parameters of Bottleneck are designed in the following table.

In the process of extracting image features, it is inevitable that some feature layers will have a large role and some feature layers will have a small role. Therefore, we use Efficient Channel Attention (ECA) [33], which is applied to the feature map by the weights extracted from the channel itself, ensuring that the channel weights are given adaptively on the basis of the features extracted from the feature map, so that the feature map with a large role has a little more influence on the results. Therefore, the final result is a little more effective in extracting features than a normal convolutional layer.

Given the aggregated features obtained by average pooling [C,1,1], the ECA module generates the channel weights by performing a one-dimensional convolution with a convolution kernel of size k, where k is determined adaptively by mapping the channel dimension C. The k is determined by:

$$K = Odd\left(\frac{log_2 C + b}{\gamma}\right) \tag{2}$$

Odd means only odd numbers can be taken, and γ is set to 2. The idea of this adaption is that when the number of channels is large, I need the convolution kernel k to be slightly larger, and when the number of channels is small, I need the convolution kernel k to be slightly smaller, so that the interaction between some of the channels can be fully fused.

Details of the structural design of Backbone and ECA are shown in the table I:

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TABLE I THE STRUCTURAL DESIGN OF BACKBONE AND ECA

Layer	Parameters	Module
Input	Shape(64×56×56)	Input Layer
Conv	1×1, c=64, k=1	Bottleneck
Batchnorm	64	
Relu	Sigmoid Function	
Conv	3×3, c=64, k=3	
Batchnorm	256	
Relu	Sigmoid Function	
Conv	1×1, 256, k=1	
Batchnorm	256	
Residual	1×1, Batchnorm, 256	
Pooling	Average pooling	ECA block
Conv	1×1, c=64, k=K, p = $\frac{k-1}{2}$	
Output	Shape(64×56×56)	Output Layer

In the above table, c represents the number of channels, k represents the kernel size, stride of the convolution process is set to 1 by default, and p represents padding. we stacked 4 Bottlenecks to form Backbone, and the number of stacking inside each Bottleneck is set to 3, 4, 6, 3 respectively. each Bottleneck is followed by Each Bottleneck is followed by ECA, and the number of preset channels of ECA is set to 64, 128, 256, 512 respectively.

B. *Dual Channel Feature Extraction Module with Convolutio -nal Block Attention Module*

In the second half of Figure 1, the remote sensing image data pass through Spatial Attention (SAM) Module, Dual Channel Feature Extraction Module (DCFM), Channel Attention Module (CAM). The SAM-DCFM-CAM architecture can solve the problem of uneven distribution of feature weights in the convolution process.

1) Spatial Multiscale Attention Module: After Backbone extracts the deepest features, We send the output into the ASKM module, which consists of the channel attention mechanism module and the selective convolution unit of SKnet. The spatial attention mechanism can extract the key spatial local features, and Sknet can extract the global features in space by adaptive multi-scale convolution.

First we calculate the spatial attention, spatial attention focuses on ”where” the effective information is on the feature map. To compute the spatial attention, we first average and maximize pooling in the channel dimension, and then concatenate (concatenate) the feature maps they generate. The final spatial attention SA feature map is then generated using a convolution operation on the stitched feature map:

$$SA = \sigma\left(f^{7\times7}\left(F^{s}_{avg};F^{s}_{max}\right)\right) \tag{3}$$

The output are then processed using the selective convolution unit in Sknet. Specifically, for the obtained features, we perform convolution operations at different scales, where there are various choices of convolution kernels, and we take the convolution kernel equal to 3 and equal to 5 as an example to perform two convolution operations respectively.

$$\begin{aligned}\overline{U} &= f^{3\times3}(SA)\\ \hat{U} &= f^{5\times5}(SA)\end{aligned} \tag{4}$$

In order to reduce the computation, here we use the more efficient group convolution, and also the null convolution, where $f^{3\times3}$ represents the convolution process with a convolution kernel of 3, and similarly, $f^{3\times3}$ represents the convolution process with a convolution kernel of 5.

$$\begin{aligned}U &= \hat{U} + \overline{U}\\ z &= \mathcal{F}_{fc}(\mathcal{F}_{gp}(\mathbf{U}))\end{aligned} \tag{5}$$

The feature maps obtained from the convolution are summed and pooled by global averaging into a $1 * 1 * C$ compressed feature, which is then reduced into Z in dimension twice to improve the processing efficiency. After that, the compressed features Z are up-dimensioned into N 1*1*C by fully connected layer, N is the number of convolution kernels.Then,the same channel with different convolution kernels and normalized to 1 by softmax . the feature maps of different scales, multiply by the corresponding final weight and then were summed to obtain the final feature maps.The formula is as follows:

$$\mathbf{V}_c = a_c \cdot \tilde{\mathbf{U}}_c + b_c \cdot \widehat{\mathbf{U}_c}, \quad a_c + b_c = 1 \tag{6}$$

C. *Training Details*

The model was developed and evaluated on a desktop computer featuring an Intel Core i7 processor running at a clock speed of 3.6 GHz, equipped with 16GB of DDR4 RAM. For the model training process, an NVIDIA GeForce RTX 2080 Ti graphics processing unit with 11GB of dedicated VRAM was utilized. The experimental environment involved operating on the Ubuntu 20.04 LTS system, with Anaconda 4.8.5 as the package manager, Python 3.7 as the programming language, and PyTorch 1.8.1 as the deep learning library. The Jupyter Notebook functioned as the environment for running the custom Python scripts, with the built-in time module for recording the execution times. The main advantage of this experimental environment was its high computational speed and efficiency, promoting faster iterations and more accurate results.

During the training process, a batch size of 32 was set, and the images were resized to 256 x 256 pixels before processing. In addition, normalization was performed using a mean and standard deviation set to [0.5, 0.5, 0.5], which assisted the model to generalize better across multiple pictures. A ResNet50 pre-trained model was loaded, and cross-entropy loss function was used in the training process. The optimizer used for training is Adamax, and the learning rate was set to 0.0001. The model was trained for 50 epochs, tracking the training and testing losses and accuracies, which were stored in lists. If the test accuracy exceeded 0.93, the model was saved. The specific parameters and settings assigned within the code were crucial in building a productive and precise image classification model.

IV. EXPERIMENTS

A. *Data Preparation*

To test the generalization ability of the recognition model more scientifically, we selected four datasets, namely SIRIWHU

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[41], RSSCN7 [30], RIS19 [34], Land of Use [13].

Each dataset precisely shows the generalization ability of the model at a certain level. They are presented separately next.

B. Baseline Models

In our research, we have selected 8 baseline models to compare with our proposed model. It is important to note that these models are all small to medium scale, and each can be deployed on microsystems. The models include: VGG19 [1],ResNet50 and ResNet18 [12],DenseNet12 [42],ResNeXt [39],ConvTExtBase [40],SceneNet [10],AcNet [6], transformer based model ViT [8] . For ViT, we research on two different sets of params namely ViT-Base(ViT-B) and ViT-Large(ViT-L).

In addition, we conducted ablation experiments using the ImageNet dataset [3], which is a classic classification benchmark. Unlike remote sensing image data, the ImageNet dataset has no deflection and displacement, and has a good sense of hierarchy within each category.

C. Classification Results

In order to further test the classification generalization performance of the model, we adjusted the ratio between the test set and the training set data, and defined the ratio as the Training Ratio value This experiment tested three sets of Training Ratio values in turn, 80%, 60%, and 40%, respectively, as shown in Table 2, 3, 4.

As shown in Table 2, under the condition that the Training Ratio value is 80% of the standard value, based on the experimental results of remote sensing image classification on four datasets, we can see that the choice of model has a significant impact on the accuracy of the classification. The highest test accuracy was achieved by our proposed model, which outperformed the other models on all four datasets: Land of Use, SIRI-WHU, RSSCN7, and RIS19. Our model achieved a test accuracy of 93.1% for Land of Use, 95.8% for SIRI-WHU, 95.1% for RSSCN7, and 94.6% for RIS19.

Among the other models, the best-performing models were ConvTextBase and SceneNet, achieving test accuracies of 92.1% and 89.8% respectively on the Land of Use dataset. ResNet50 and ResNet18 also performed well on SIRI-WHU dataset, with test accuracies of 92.7% and 93.1% respectively. DenseNet121 performed well on the RSSCN7 dataset, with a test accuracy of 94.6%.

VGG19 model had the lowest performance on all four datasets, with test accuracies of 60.7% on Land of Use, 89.6% on SIRI-WHU, 87.3% on RSSCN7, and 89.6% on RIS19. Experiments show that the ViT model with large params slightly outperforms our model. However, the complexity of our model is lower than these ViT models with large params (demonstrated in Figure 3).

TABLE II EXPERIMENTAL RESULTS ON FOUR DIFFERENT DATASETS(TRAINING RATIO=80%).

Training Ratio=80%		**Test Accuracy(%)**		
Model	**Land of Use**	**SIRI-WHU**	**RSSCN7**	**RIS19**
VGG19 [1]	60.7	89.6	87.3	89.6
Resnet50 [12]	88.6	92.7	88.1	86.4
Resnet18 [12]	85.2	93.1	86.7	92.6
Densenet121 [42]	60.7	94.6	91	90.8
Resnext [39]	90.1	93.6	87.3	88.6
Convtextbase [40]	92.1	95.4	91.6	91.7
SceneNet [10]	89.8	95.1	94.2	94.3
ViT-B [8]	72.4	84.6	85.2	90.1
ViT-L [8]	91.5	92.7	98.5	90.0
Blur-ASKNet(Ours)	93.1	95.8	95.1	94.6

TABLE III EXPERIMENTAL RESULTS ON FOUR DIFFERENT DATASETS(TRAINING RATIO=60%).

Training Ratio=60%		**Test Accuracy(%)**		
Model	**Land of Use**	**SIRI-WHU**	**RSSCN7**	**RIS19**
VGG19 [1]	59.1	87.4	90.7	89.6
Resnet50 [12]	83.9	89.7	82.4	85.9
Resnet18 [12]	75.3	86.7	90.2	88.4
Densenet121 [42]	56.2	82.1	83.4	88.8
Resnext [39]	90.1	93.6	87.3	79.6
Convtextbase [40]	87.0	90.2	83.0	93.1
SceneNet [10]	83.7	75.3	90.4	85.0
ViT-B [8]	79.2	82.1	87.4	87.1
ViT-L [8]	85.4	95.3	94.2	93.9
Blur-ASKNet(Ours)	93.2	96.4	92.4	95.0

TABLE IV EXPERIMENTAL RESULTS ON FOUR DIFFERENT DATASETS(TRAINING RATIO=40%).

Training Ratio=80%	**Test Accuracy(%)**			
Model	**Land of Use**	**SIRI-WHU**	**RSSCN7**	**RIS19**
VGG19 [1]	70.5	79.2	80.3	77.1
Resnet50 [12]	79.5	84.4	87.5	79.8
Resnet18 [12]	83.9	90.4	82.5	91.9
Densenet121 [42]	64.7	87.3	86.3	91.4
Resnext [39]	85.7	90.5	81.8	85.4
Convtextbase [40]	93.8	89.4	92.9	90.6
SceneNet [10]	82.3	85.4	88.9	79.9
ViT-B [8]	76.1	80.9	86.6	88.5
ViT-L [8]	92.3	91.9	89.7	94.5
Blur-ASKNet(Ours)	90.4	94.5	90.4	92.1

And when we turn down the Training Ratio value in turn, the accuracy of each model classification slightly decreases,as shown in Table 3, 4, but the recognition rate of our model has been maintained at the highest level, stronger than the state-of-the-art methods.

To show the classification ability of the model more intuitively, we used a confusion matrix to visualize the results, and this paper shows the classification on the SIRI-WHU dataset at Training Ratio=80%.

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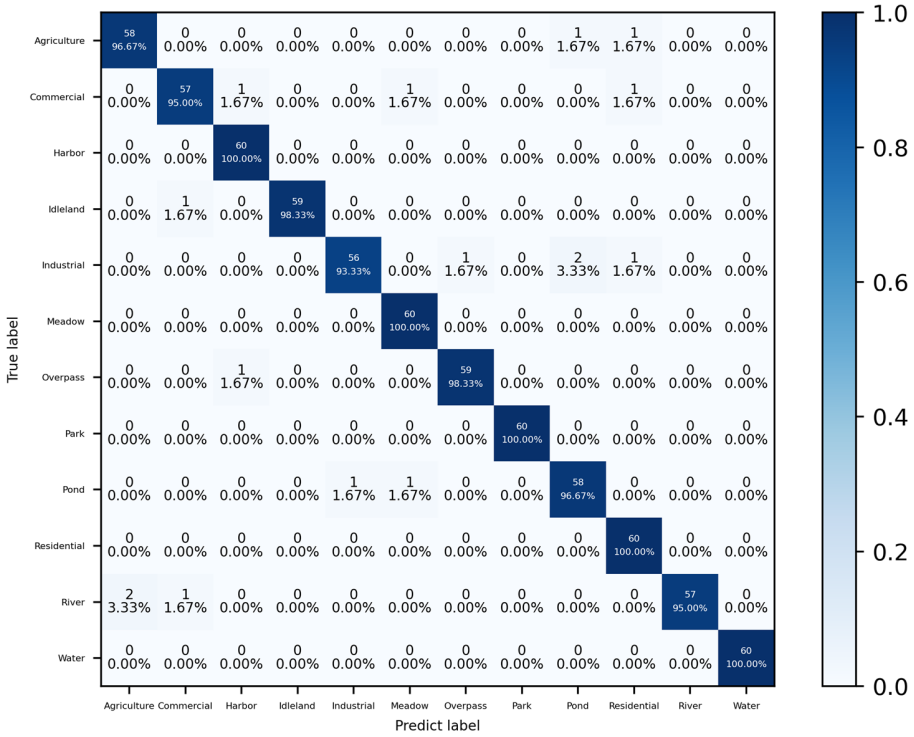


Fig. 2. Confusion Matrix in SIRI-WHU Dataset

In Figure 2, very few commercial, harbor, and pond images have false positives, but the recognition rates of all images are above 95 %, indicating that the overall performance of our model is excellent when Training Ratio=80 %.

D. Ablation Study

To validate the respective roles of each of our proposed modules, we use ECA as the baseline model for this experiment, which replaces blur-pooling (BL) with traditional max pooling without the inclusion of SAM and Sknet. this experiment is divided into a total of five groups and tested on four different datasets, with the following results where Training Ratio is set to be 80% and 60% respectively, as shown in Table 5, 6:

TABLE V ABLATION STUDY RESULTS ON FOUR DIFFERENT DATASETS.

Training Ratio=80%	**Test Accuracy(%)**			
Model	**Land of Use**	**SIRI-WHU**	**RSSCN7**	**RIS19**
ECA [33]	70.1	75.5	69.1	68.4
ECA+BL	73.5	76.7	72.2	74.1
ECA+BL+SAM	80.4	84.1	89.2	87.9
ECA+BL+Sknet	86.5	89.9	83.5	82.0
ECA+BL+SAM+Sknet	89.7	92.5	94.1	88.9

From the two groups of experiments, ECA [33] and ECA+BL, it can be seen that the classification performance of the network without BL processing is significantly worse than that of the ECA network with BL added because most of the remotely sensed images have rotation angle, and the introduction of both SAM and Sknet can effectively improve the classification performance of the network. The combination of the two can substantially improve the correct rate of model recognition, which indicates that the two are less coupled with each other and have some independence from each other in the logic of optimization, so they can collaborate to improve the recognition ability of the model.

Further, we tested the performance of two groups, ECA [33], ECA+BL, on Imagenet by setting three Training Ratio values of 40 %, 60 %, 80 %, respectively..

TABLE VI ABLATION STUDY RESULTS ON IMAGENET DATASET

Model	**Test Accuracy(%) TR(Training Ratio)=80%**	**TR=60%**	**TR=40%**
ECA [33]	74.7	85.5	82.4
ECA+BL	73.9	87.9	79.4

The recognition rates of ECA [33], ECA+BL in the Imagenet dataset are not very different, but similar to those in the remote sensing dataset. The addition of BL even brings some fluctuation to the recognition rate, indicating that the applicability of our proposed model is not strong on a standard dataset like Imagenet, which is not remotely sensed. It also shows that our idea focuses specifically on the field of remote sensing image classification and is somewhat migratory.

E. Model Complexity Analysis

In classification networks, the increase of recognition rate may bring about the problems of increased model complexity, increased number of parameters, and longer training time. To verify whether this ”side effect” is within the acceptable range, we compare our model with Resnext, Convtextbase, and SceneNetzuo, which have relatively high recognition rates in comparison experiments, and compare the number of parameters and training time of the models, respectively.

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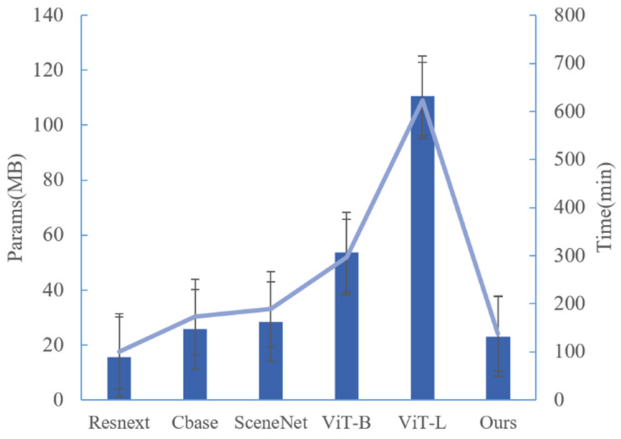


Fig. 3. Model Complexity Analysis

As shown in Figure 3, the number of model parameters and the training time are positively proportional in the overall, the longer the training time, the higher the number of parameters. With recognition rates higher than the Resnext, Cbase, SceneNet, ViT-L, our model counts and training times are higher than those of Resnext but lower than those of Convtexbase and SceneNet. Although ViT-L model performs higher accuracy, but its scale are far more than our proposed model. This indicates that our model reconciles well the relationship between model counts and training times.

V. Conclusions

This academic paper aims to address two major challenges in remote sensing image classification problem, the intraclass variability and inter-class similarity. To overcome these challenges, we propose Blur-ASKM, which employs rotation undeformation to achieve blurpooling. Blur-ASKM not only solves the intra-class variability issue but also introduces ASKM to obtain joint features, thus tackling the problem of computational cost and number of model parameters. Furthermore, this approach significantly improves the efficiency of small and medium-sized recognition model. For future work, we intend to investigate the effectiveness of our ASKM idea in large models such as Transformer-based models, and optimize the multi-scale fusion and adaptive algorithms to further enhance the model performance. The ultimate goal is to develop a larger and stronger model that can efficiently solve classification issues encountered in remote sensing image data.

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